

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

MI-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Surface Mining Planning

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data wherever necessary.

1. a) Discuss the safety and constructional features of good and efficient haul roads for open pit mining.
b) What is balancing diagram for Dragline? Explain. How the stacking height and dumping radius of dragline is calculated from this diagram? Derive the expression.
2. a) Explain vertical tandem operation of Dragline by drawing suitable layout.
b) Describe a Bucket wheel excavator with a neat sketch under the following headings:
 - i) Constructional features
 - ii) Wheel design
3. a) Write short notes on:
 - i) Continuous Mining method
 - ii) Dropping Cut method
b) Describe constructional features of Continuous Surface Miner with suitable diagram.

MI-7004(2)-CBGS

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[2]

4. a) What is floating cone technique? Explain step by step. Determine the ultimate pit size for the following economic block model using the above technique. Also determine the NAV of the optimum pit and the overall stripping ratio.

-1	-2	-1	+2	-1	-1	-1
-1	-1	-1	+4	+2	-1	-2
-2	-3	-1	+6	-1	-2	-1

- b) i) How economic block model is prepared for a mineral deposit? Explain.
- ii) Explain breakeven stripping ratio and its significance.
5. a) Write down the objectives of production planning and role of cut off grade in open pit production planning and scheduling. Determine the value per tonne of copper ore and the breakeven stripping ratio for the following data:
Grade - 3% Cu,
Overall recovery - 90%
Total cost of production of Mineral - Rs. 5000/ Tonne of ore
Selling Price - Rs. 300/kg of Cu
Cost of Stripping - Rs. 70/m³
- b) Describe the various geometrical conditions under which wedge and toppling failure occurs.
6. a) Explain the techniques used for analysis of slopes under circular failure condition.
- b) Discuss the various geotechnical factors influencing the stability of hard rock slopes.

[3]

7. a) Write notes on:
 - i) Slope monitoring
 - ii) Finite element method
 - b) Discuss step by step the procedure for carrying out slope stability investigation of a large open pit mine in a geologically disturbed area.
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8. a) Discuss the various techniques of slope stabilization in brief.
 - b) A 60m high slope has an overall slope angle of 50° made from three 20m high benches with 70° slope. The slope is in reasonable fresh granite with several sets of steeply dipping joints. The slope is an area of high rainfall and low seismicity with an acceleration of $0.08g$. On the basis of structural mapping it has been found that sheet jointing with 35° dip may cause plane failure along it. The friction angle and cohesion was found 30° and 150kPa along this surface. The density of rock and water is 0.027 mN/m^3 and 0.01 mN/m^3 . The depth of tension cracks is 14m. Assess the stability of slope, when the tension crack is completely dry.
